Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIFTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022

Course Code: EET 305

Course Name: SIGNALS AND SYSTEMS

PART A

Answer all questions. Each question carries 3 marks

Q1 Plot $x(2t)$, if $x(t)=u(-t+2)+u(t-3)$.

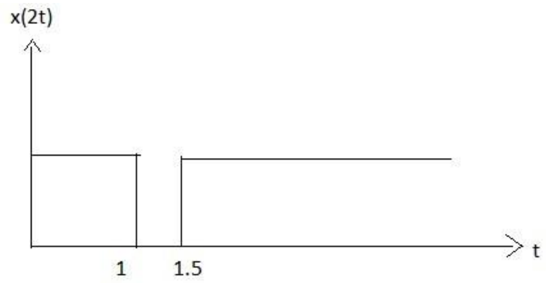
Ans To plot $x(2t)$, we need to substitute $2t$ for t in the given expression for $x(t)$:

$x(2t) = u(-2t+2) + u(2t-3)$

The first unit step function $u(-2t+2)$ is activated when $-2t+2 \geq 0$, which is equivalent to $t \leq 1$.

The second unit step function $u(2t-3)$ is activated when $2t-3 \geq 0$, which is equivalent to $t \geq 3/2$.

Therefore, $x(2t)$ will be nonzero only in the interval [0.5, 1.5]. Within this interval, both unit step functions are active, and we have:



Q2 Find whether following signal is odd or even, $x(t)=t4+4t2+6$.

Ans To determine whether the signal $x(t)$ is odd or even, we need to check if it satisfies the conditions of odd or even functions.

An even function is defined as $f(-x)=f(x)$ for all x.

An odd function is defined as $f(-x)=-f(x)$ for all x.

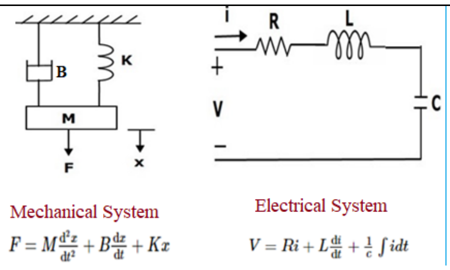
Let's first check for evenness:

$x(-t) = (-t)^4 + 4(-t)^2 + 6$

$= t^4 + 4t^2 + 6$ (since t^4 and t^2 are even functions)

Since $x(t) = x(-t)$, $x(t)$ is an even function.

	Q3	State and prove the time shifting property of Fourier Transform.
	Ans	The time-shifting property of the Fourier transform states that if we shift a function in the time domain by a certain amount, then its Fourier transform will also be shifted by the same amount in the frequency domain. Proof: We start with the definition of the Fourier transform: $F\{f(t)\} = \int f(t)e^{(-i\omega t)}\,dt$ Now, let us consider the time-shifted function $g(t) = f(t - \tau)$. Using the substitution $u = t - \tau$, we can rewrite the Fourier transform of g(t) as: $F\{g(t)\} = \int g(t)e^{(-i\omega t)}\,dt = \int f(u - \tau)e^{(-i\omega(u+\tau))}\,du$ Next, we use the property that $e^{(-i\omega\tau)}$ is a constant with respect to the variable of integration u, so we can factor it out of the integral: $F\{g(t)\} = e^{(-i\omega\tau)}\int f(u)e^{(-i\omega u)}\,du$ Finally, we recognize that the integral on the right-hand side is simply the Fourier transform of f(t), so we obtain: $F\{g(t)\} = e^{(-i\omega\tau)}F\{f(t)\}$
	Q4	Derive the electrical analogous elements corresponding to elements of mechanical translational system with force voltage analogy.
	Ans	In electrical engineering, the force-voltage analogy is often used to model mechanical systems as electrical circuits. In this analogy, the mechanical quantities such as force, velocity, displacement, and mass are represented by their electrical equivalents: voltage, current, charge, and capacitance/inductance, respectively. Let us consider a mechanical translational system consisting of a mass m, a spring with spring constant k, and a damping coefficient b, which is subjected to an external force F(t). We can derive the electrical analogous elements corresponding to the elements of this mechanical system as follows: Mass (m): The electrical equivalent of mass is capacitance. The capacitance C is given by $C = m/k$, where k is the spring constant. Spring (k): The electrical equivalent of spring is inverse capacitance or inductance. The inductance L is given by $L = 1/k$. Damping (b): The electrical equivalent of damping is resistance. The resistance R is given by $R = b$. External force (F(t)): The electrical equivalent of external force is voltage source. The voltage source V(t) is given by $V(t) = F(t)/C$. Using these electrical analogues, we can represent the mechanical translational system as an electrical circuit, as shown in the figure below:



Here, the capacitor C represents the mass, the inductor L represents the spring, and the resistor R represents the damping. The voltage source V(t) represents the external force F(t).

force-voltage analogy are: capacitance for mass, inductance for spring, resistance for damping, and voltage source for external force.

Q5 Explain the concept of positive real functions.

Ans A function is said to be positive real if it satisfies two properties:

It is a real-valued function that is defined over the positive real numbers (i.e., the domain of the function is the set of all positive real numbers).

It satisfies a specific condition called the positive-real condition, which relates to the frequency response of the function.

The positive-real condition states that a function H(s) is positive real if and only if the real part of H(jω) is non-negative for all ω, where j is the imaginary unit ($j^2 = -1$) and ω is the frequency. In other words, the function H(s) is positive real if its frequency response has no phase shift greater than 90 degrees and its magnitude is always non-negative.

Q6 Determine the impulse response of the system having transfer function, $H(s)=s\,/(s-2)(s+3)$

Ans To determine the impulse response of the system, we need to take the inverse Laplace transform of the transfer function H(s) = s / ((s-2)(s+3)):

H(s) = s / ((s-2)(s+3))

= A/(s-2) + B/(s+3) + C(s)

Multiplying both sides by the denominator (s-2)(s+3) and simplifying, we get:

s = A(s+3) + B(s-2) + C(s-2)(s+3)

Setting s = 2, we get:

2 = 5A - 2B

Setting s = -3, we get:

-3 = A - 9B

Setting s = 0, we get:

		$0 = -6C$ Therefore, $C = 0$. Solving for A and B, we get $A = -3/7$ and $B = 2/7$. So we can write: H(s) = (-3/7)/(s-2) + (2/7)/(s+3) Taking the inverse Laplace transform of each term, we get: h(t) = (-3/7)e^(2t)u(t) + (2/7)e^(-3t)u(t) where u(t) is the unit step function. Therefore, the impulse response of the system is: h(t) = (-3/7)e^(2t)u(t) + (2/7)e^(-3t)u(t)
	Q7	Define the process of sampling and state sampling theorem.
	Ans	Sampling is the process of converting a continuous-time signal into a discrete-time signal. This process involves taking samples of the continuous-time signal at regular intervals of time and converting them into discrete values. The Sampling Theorem, also known as the Nyquist-Shannon Sampling Theorem, states that a continuous-time signal can be reconstructed from its samples, provided that the sampling rate is at least twice the highest frequency component in the signal. In other words, the sampling rate must be greater than or equal to the Nyquist rate, which is equal to twice the highest frequency present in the signal. The process of sampling involves the following steps: Sampling: The continuous-time signal is sampled at regular intervals of time. Quantization: The continuous amplitude values of the signal are approximated by a finite set of discrete values. Encoding: The discrete values are encoded using a binary code. The resulting discrete-time signal can then be processed using digital signal processing techniques such as filtering, modulation, and demodulation. In order to reconstruct the original continuous-time signal, the discrete-time signal can be passed through an interpolation filter, which will remove any spurious frequency components and recover the original signal.
	Q8	Write any three properties of z-transform.
	Ans	The z-transform is a mathematical technique used in digital signal processing to transform a discrete-time signal into a complex frequency-domain representation. Here are three properties of the z-transform: Linearity: The z-transform is a linear transformation, which means that if you have two discrete-time signals x[n] and y[n] and constants a and b, then the z-transform of ax[n] + by[n] will be aX(z) + bY(z), where X(z) and Y(z) are the z-transforms of x[n] and y[n], respectively.

		Time-shifting: The z-transform has a time-shifting property, which means that if you shift a discrete-time signal x[n] by a certain amount k, then the z-transform of x[n-k] will be z^(-k)X(z), where X(z) is the z-transform of x[n]. Convolution: The z-transform has a convolution property, which states that the z-transform of the convolution of two discrete-time signals x[n] and y[n] is equal to the product of their individual z-transforms X(z) and Y(z). In other words, if z-transforms of x[n] and y[n] are X(z) and Y(z) respectively, then z-transform of the convolution x[n]*y[n] is X(z)Y(z).
	Q9	Check the stability of the system represented by the following characteristic equation using Jury's test: $2z4+7z3+10z2+4z+1=0$
	Ans	To apply Jury's stability criterion, we need to rewrite the characteristic equation in the following form: $a_n z^n + a_{n-1} z^{n-1} + ... + a_1 z + a_0 = 0$ where a_i are the coefficients of the characteristic equation. In this case, the given characteristic equation is: $2z^4 + 7z^3 + 10z^2 + 4z + 1 = 0$ which can be written as: $2z^4 + 7z^3 + 10z^2 + 4z + 1 = a_4 z^4 + a_3 z^3 + a_2 z^2 + a_1 z + a_0$ The first step of Jury's stability test is to form the first row of the Jury table, which is simply the coefficients of the characteristic equation in reverse order: $1\ \ 4\ \ 10\ \ 7\ \ 2$ The remaining rows of the table are formed using the following recursion: $c_{2k} = a_{n-k}$ $c_{2k+1} = c_{2k} - a_k c_{2k-1}$ Using this recursion, we can form the remaining rows of the table as follows: $1\ \ 4\ \ 10\ \ 7\ \ 2$ $4\ \ 6\ \ -3\ \ -6$ $6\ \ -15\ \ -21$ $-15\ \ -141$ -156 According to Jury's stability criterion, the system represented by the given characteristic equation is stable if and only if all the elements of the first column of the table have the same sign All the elements have different signs, so the system is unstable. Therefore, the given system is unstable.
	Q10	Obtain the transfer function of given discrete time system described by the difference equation: $y(n)-5/4y(n-1)+1/6y(n-2)=2x(n)$
	Ans	The given difference equation is:

$y(n) - 5/4y(n-1) + 1/6y(n-2) = 2x(n)$

Taking the Z-transform of both sides, using the property of linearity and the time-shifting property, we get:

Y(z) - 5/4 z^(-1) Y(z) + 1/6 z^(-2) Y(z) = 2 X(z)

Simplifying and solving for the transfer function H(z) = Y(z) / X(z), we get:

H(z) = Y(z) / X(z) = 2 / (1 - 5/4 z^(-1) + 1/6 z^(-2))

Multiplying both the numerator and denominator by z^2, we get:

H(z) = (2 z^2) / (z^2 - 5/4 z + 1/6)

Thus, the transfer function of the given discrete-time system is:

H(z) = (2 z^2) / (z^2 - 5/4 z + 1/6)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

Q11 For the network given below,

a) Find the response of a system with an impulse response of $h(t)= u(t)$ for an input signal of $x(t)=e{-}{\propto}t\ u(t)$ (8)

b) Determine whether the following systems are time-invariant and linear.

(i) $3 d\,y(t)/dt + 5ty(t) = x(t)$

(ii) $d\,y(t)/dt + 10\sin y(t) = 2x(t)$

(iii) $y(t) d\,y(t)/dt + 10\ y(t) = 2x(t)$

Ans

a) The output response of a system can be calculated by convolving the input signal with the impulse response of the system.

In this case, the impulse response of the system is $h(t) = u(t)$, which is the unit step function.

The input signal is $x(t) = e$^$(-\propto t)\ u(t)$, which is a decaying exponential function multiplied by the unit step function.

So, the output response of the system can be found as:

$y(t) = x(t) * h(t) = ∫\_(-\infty)$^$\infty\ x(\tau)\ h(t-\tau)$ dτ

Since $h(t) = u(t)$, we can simplify the above expression as:

$y(t) = ∫\_0$^$t\ x(\tau)$ dτ

Substituting $x(t) = e$^$(-\propto t)\ u(t)$, we get:

$y(t) = ∫\_0$^$t\ e$^$(-\propto\tau)\ u(\tau)$ dτ

Using integration by substitution, we can simplify the above expression as:

$y(t) = [-e$^$(-\propto\tau)/\propto]\_0$^$t = [-e$^$(-\propto t)/\propto] + [e$^$0/\propto] = (1/\propto)\ (1 - e$^$(-\propto t))$

Therefore, the output response of the system is $y(t) = (1/\propto)\ (1 - e$^$(-\propto t))$.

b) (i) $3 d\,y(t)/dt + 5ty(t) = x(t)$

To determine whether the system described by the differential equation:

$3dy(t)/dt + 5ty(t) = x(t)$ is time-invariant and linear, we need to check if it satisfies the following properties:

Homogeneity (scaling property): If we apply an input $ax(t)$, the output of the system will be:
$3d(ay(t))/dt+5t(ay(t)) = a(3dy(t)/dt+5ty(t)) = ay(t)$
Thus, the homogeneity property holds.

Additivity (superposition property): If we apply two inputs $x1(t)$ and $x2(t)$, the corresponding outputs will be $y1(t)$ and $y2(t)$, respectively. Then, the output corresponding to the input $x1(t) + x2(t)$ will be:
$3d(y1(t)+y2(t))/dt+5t(y1(t)+y2(t)) = 3(dy1(t)/dt+dy2(t)/dt) + 5ty1(t)+5ty2(t)$

By using the linearity property of the derivative, we can write $d(y1(t)+y2(t))/dt = dy1(t)/dt+dy2(t)/dt$.

Therefore the signal is Linear Time varying

(ii) $d\,y(t)/dt+10\sin y(t)=2x(t)$

Homogeneity (scaling property): If we apply an input $ax(t)$, the output of the system will be:
$d(ay(t))/dt+10\sin(ay(t))=2ax(t)$
We can see that the equation is not linear in x, since the term $\sin(ay(t))$ is not linear. Thus, the system is not linear.

Now check time-invariance:

If we apply the input $x(t)$, the output of the system will be $y(t)$. If we apply the input $x(t-t0)$, the output of the system will be $y(t-t0)$. Thus, the output is shifted in time by $t0$.

Let's see if the output $y(t-t0)$ can be obtained by applying the input $x(t-t0)$ to the system:

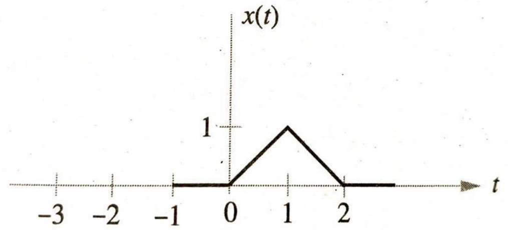
$d\ y(t-t0)/dt+10\sin y(t-t0)=2x(t-t0)$

This equation is not the same as the original equation $d\ y(t)/dt+10\sin y(t)=2x(t)$, since the argument of sin in the second equation is t, while in the first equation it is $t-t0$. Thus, the system is not time-invariant.

(iii) $y(t)d\,y(t)/dt+10\ y(t)=2x(t)$

Homogeneity (scaling property): If we apply an input $ax(t)$, the output of the system will be:
$ay(t)d(ay(t))/dt+10\ ay(t)=2ax(t)$
We can see that the equation is linear in x and y, since all terms are linear. Thus, the system is linear.

Now let's check time-invariance:

		If we apply the input $x(t)$, the output of the system will be $y(t)$. If we apply the input $x(t-t0)$, the output of the system will be $y(t-t0)$. Thus, the output is shifted in time by $t0$. Let's see if the output $y(t-t0)$ can be obtained by applying the input $x(t-t0)$ to the system: $y(t-t0)d\ y(t-t0)/dt+10\ y(t-t0)=2x(t-t0)$ Therefore the signal is Non linear time varying
	Q12	For the signal, $x(t)$ shown , plot: (i) $x(t-1)$ (ii) $x(-3t+2)$
	Ans	 To plot $x(t-1)$, we need to shift the original signal $x(t)$ to the right by 1 unit of time. This can be done by replacing t with $t+1$ in the original signal $x(t)$, which gives $x(t-1)=u(t+1)-u(t-1)$, where $u(t)$ is the unit step function. To plot $x(-3t+2)$, we need to compress the original signal $x(t)$ by a factor of 3 and then shift it to the right by 2 units of time. This can be done by replacing t with $(t-2)/3$ in the original signal $x(t)$, which gives $x(-3t+2)=u((t-2)/3)-u((t-8)/3)$. The amplitude of the compressed signal is the same as the original signal, but the width of the signal is compressed by a factor of 3. Also, the signal is shifted to the right by 2 units of time
		(b) Test whether the following signals are periodic or not. If periodic, find the fundamental period. (i) $x(t)=5\sin 24\pi t+7\sin 36\pi t$ To determine if a continuous-time signal is periodic or not, we need to check whether it repeats itself after some fixed time interval, called the period. Let's consider the signal $x(t)=5\sin(24\pi t)+7\sin(36\pi t)$. To test if the signal is periodic, we need to find a time interval T such that $x(t)=x(t+\mathrm{T})$ for all values of t. We know that sin(θ) is periodic with a period of 2π, which means that sin(θ)=sin(θ+2πk) for all integer values of k. Using this property, we can rewrite the signal as: $x(t) = 5\sin(24\pi t) + 7\sin(36\pi t)$ $= 5\sin(2\pi(12t)) + 7\sin(2\pi(18t))$ We can see that the first term has a period of T1=1/12 seconds and the second term has a period of T2=1/18 seconds. Therefore, the signal $x(t)$ is periodic

(ii) $x(t)$=$ej(\pi t-2)$

To test whether the signal $x(t)$=e^($j(\pi t-2)$) is periodic or not, we need to find a value of T such that $x(t)$=$x(t$+T) for all values of t.
The only way this equation can hold for all values of t and k is if πT is an integer multiple of 2π, i.e., πT=2πn for some integer n. Thus, the fundamental period of the signal is:

T = 2n

Since n can be any integer, the signal $x(t)$=e^($j(\pi t-2)$) is not periodic.

(iii) $x(t)=5\sin 24\pi t\cos 36\pi t$

To test whether the signal $x(t)$=5sin(24πt)cos(36πt) is periodic or not, we need to find a value of T such that $x(t)$=$x(t$+T) for all values of t.

We can use the product-to-sum formula to rewrite $x(t)$ as:

$x(t)$ = 2.5[sin(60πt) + sin(12πt)]

Let's assume that the signal is periodic with period T, so we have:

$x(t)$ = $x(t$+T)

2.5[sin(60πt) + sin(12πt)] = 2.5[sin(60π(t+T)) + sin(12π(t+T))]

sin(60πt) + sin(12πt) = sin(60πt+60πT) + sin(12πt+12πT)

Using the sum-to-product formula, we can simplify the right-hand side of the equation above as:

sin(60πt+60πT) + sin(12πt+12πT) = 2sin(36πt+36πT)cos(24πT)

Substituting this into the equation above, we get:

sin(60πt) + sin(12πt) = 2sin(36πt+36πT)cos(24πT)

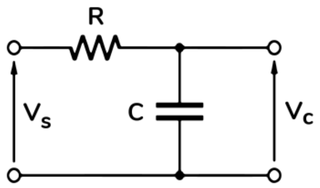
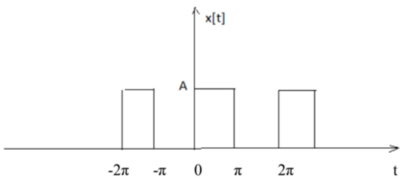
Dividing both sides by cos(24πT), we get:
tan(60πt) + tan(12πt) = 2tan(36πt+36πT)

Taking the inverse tangent of both sides, we get:
60πt + 12πt = 36πt + 36πT + $n\pi$, where n is an integer.

Simplifying the equation above, we get:

T = (n/6) - (2/3)

Since T depends on the value of n, which can be any integer, the signal $x(t)$=5sin(24πt)cos(36πt) is not periodic

		Module 2
	Q13	Determine the transfer function, $VC(s)$ / $Vs(s)$ of the given circuit. Hence, find the impulse response of the circuit. 
	Ans	$Z_R = R$ $Z_C = \frac{1}{sC}$ $V_{out} = V_{in}\frac{Z_C}{Z_C + Z_C + Z_R}$ $V_{out} = V_{in}\frac{Z_C}{2Z_C + Z_R}$ $V_{out} = V_{in}\frac{\frac{1}{sC}}{2\frac{1}{sC} + R}$ $V_{out} = V_{in}\frac{1}{2 + sCR}$ $V_{out} = V_{in}\frac{\frac{1}{RC}}{2\frac{1}{RC} + s}$ $\frac{V_{out}}{V_{in}} = \frac{\frac{1}{RC}}{2\frac{1}{RC} + s}$
	Q14	Obtain the trigonometric Fourier Series of the given signal 
	Ans	To obtain the trigonometric Fourier series of a periodic signal, we need to calculate its Fourier coefficients, which are given by the following equations:

$$a_0 = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)dt$$

$$a_n = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)\cos(n\omega_0 t)dt$$

$$b_n = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)\sin(n\omega_0 t)dt$$

where T is the fundamental period of the signal, $\omega_0 = \frac{2\pi}{T}$ is its fundamental frequency, and n is an integer.

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{2}t\right) + b_n \sin\left(\frac{n\pi}{2}t\right)\right]$$

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)\,dt$$

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(n\Omega_0 t)\,dt$$

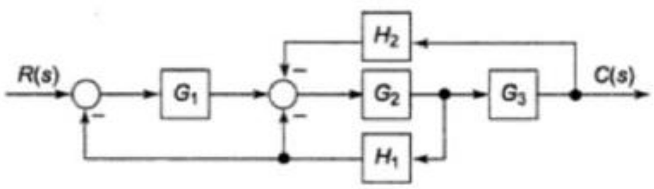
$$b_n = \frac{4}{n\pi}\sin\left(\frac{n\pi}{2}\right).$$

Module 3

Q15

a) How time response can be obtained from pole zero plot.

b) Describe Mason's gain formula for signal flow graph. Obtain the transfer function of the following system using Mason's gain formula.



Ans

a) To obtain the time response from a pole-zero plot, you need to perform the following steps:

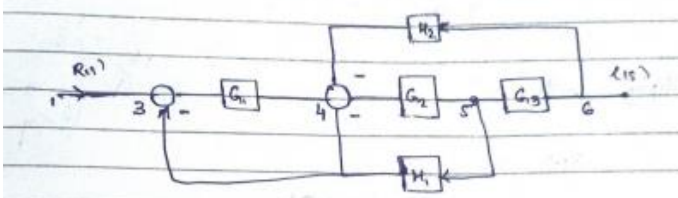
Determine the transfer function of the system from the pole-zero plot. The transfer function is the ratio of the output of the system to the input.

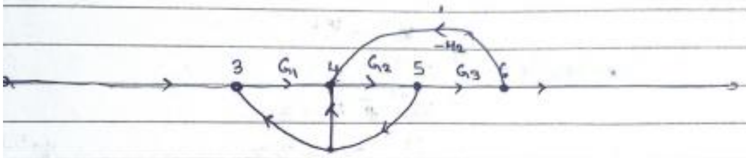
Using the transfer function, calculate the impulse response of the system. This can be done by taking the inverse Laplace transform of the transfer function.

Once you have the impulse response, you can use it to determine the time response of the system for any input signal. The time response can be found by convolving the input signal with the impulse response.

b)

$s^2 Y(s) + 3s.Y(s) + 2Y(s) = 2sX(s) - 3X(s).$





Forward path k = 1.

no possible combination of 2 non

Calculation Δ & Δ_k

$\Delta = 1 - [P_{11} + P_{21} + P_{31}]$

$= 1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1$

$\cancel{\Delta_2 = 1 - 0}$

$T = \frac{1}{\Delta} \sum_k P_k \Delta_k = \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2]$

$= \frac{1}{\Delta} [G_1 G_2 G_3 + [1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1)]$

$= \frac{G_1 G_2 G_3 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1 + 1}{1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1}$

	Q16	Check if the system with the given characteristic equation is stable using Routh Hurwitz stability criterion, $s4+8s3+18s2+16s+5=0$
	Ans	The Routh-Hurwitz stability criterion can be used to determine the stability of a system by examining the signs of the coefficients of the characteristic equation. The characteristic equation for the system in this case is: s^4 + 8s^3 + 18s^2 + 16s + 5 = 0 To apply the Routh-Hurwitz criterion, we need to create a Routh array as follows: 1 18 5 8 16 0 31/4 5 5 The first two rows of the array are obtained directly from the coefficients of the characteristic equation. The subsequent rows are calculated from the previous rows according to the following formulas: For the (n+1)-th row: The first element is the coefficient of s^n in the characteristic equation. The second element is the coefficient of s^(n-1) in the characteristic equation. The remaining elements are calculated using the following formula: (a[n-1]*a[n+1] - a[n]*a[n-2])/a[n-1] where a[n] is the coefficient of s^n in the characteristic equation. Examining the signs of the elements in the first column of the Routh array, we see that they are all positive, which is a necessary condition for stability. Next, we examine the signs of the elements in the second column of the Routh array. We see that the signs are also all positive, which means that the system is stable.

		Therefore, we can conclude that the system with the given characteristic equation s^4+8s^3+18s^2+16s+5=0 is stable according to the Routh-Hurwitz stability criterion.
		Module 4
	Q17	Explain the significance of Zero order hold circuit in signal reconstruction.
	Ans	In signal processing, the Zero-Order Hold (ZOH) circuit is an important element for reconstructing analog signals from their digital samples. The ZOH circuit holds the value of the digital sample constant for the duration of the sampling interval, effectively creating a step function that represents the value of the sample during that interval. The significance of the ZOH circuit in signal reconstruction is that it helps to prevent distortion and aliasing, which are common problems in digital signal processing. Distortion occurs when the original signal is not accurately represented by the digital samples, while aliasing occurs when high-frequency components of the signal are incorrectly interpreted as lower frequencies. The ZOH circuit ensures that the reconstructed analog signal is smooth and continuous, with no abrupt changes in amplitude between adjacent samples. This is important for many applications, such as audio processing and control systems, where a smooth and continuous signal is necessary for accurate and reliable performance. In addition, the ZOH circuit has a simple design and is easy to implement, making it a popular choice for signal reconstruction in practical applications. It is often used in conjunction with other techniques, such as interpolation and filtering, to further improve the accuracy and quality of the reconstructed signal. Overall, the Zero-Order Hold circuit is an important element in signal reconstruction as it helps to ensure that the reconstructed analog signal is accurate, continuous, and free from distortion and aliasing
	Q18	a) State and prove the time reversal property of Discrete time Fourier transform. b) Find the inverse Z-transform of: $X(Z) = \frac{z^{-1}}{3 - 4z^{-1} + z^{-2}}; ROC\|Z\| > 1$

Ans

a) The Discrete-time Fourier Transform (DTFT) is a mathematical tool used to analyze the frequency content of discrete-time signals. It converts a sequence of discrete samples into a continuous, periodic function of frequency.

The DTFT of a discrete-time signal x[n] is defined as:

X(e^(jω)) = Σ from -∞ to ∞ x[n] e^(-jωn)

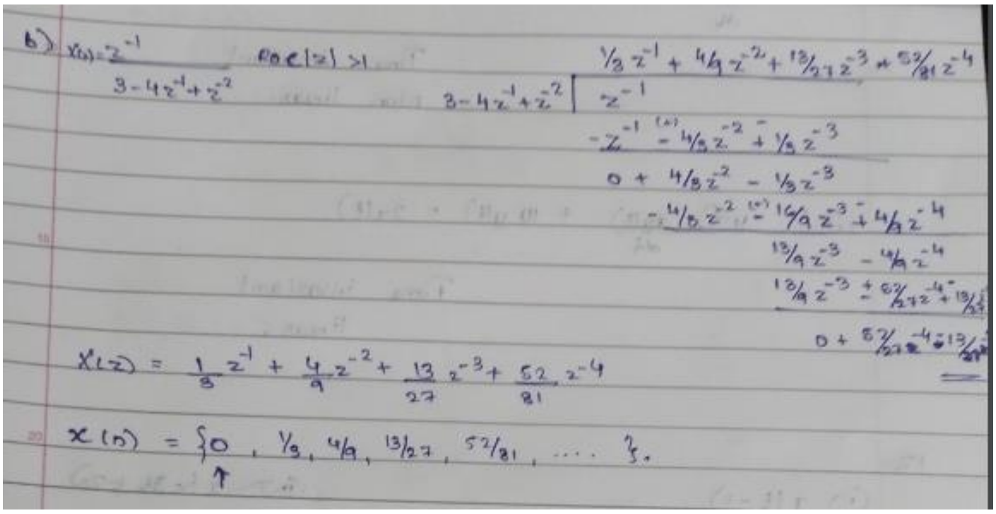
where j is the imaginary unit, ω is the normalized frequency in radians/sample, and X(e^(jω)) is a complex-valued function of ω.

To prove the DTFT, we first need to define the inverse DTFT (IDTFT):

x[n] = (1/2π) ∫ from -π to π X(e^(jω)) e^(jωn) dω

where the integral is over the range of all possible frequencies.

b)



b) $X(z) = \frac{z^{-1}}{3-4z^{-1}+z^{-2}}$ ROC $|z|>1$

$$\frac{1}{3}z^{-1} + \frac{4}{9}z^{-2} + \frac{13}{27}z^{-3} + \frac{52}{81}z^{-4}$$

$3-4z^{-1}+z^{-2} \,\big)\, z^{-1}$

$-z^{-1} - \frac{4}{3}z^{-2} + \frac{1}{3}z^{-3}$

$0 + \frac{4}{3}z^{-2} - \frac{1}{3}z^{-3}$

$-\frac{4}{3}z^{-2} - \frac{16}{9}z^{-3} + \frac{4}{9}z^{-4}$

$\frac{13}{9}z^{-3} - \frac{4}{9}z^{-4}$

$$X(z) = \frac{1}{3}z^{-1} + \frac{4}{9}z^{-2} + \frac{13}{27}z^{-3} + \frac{52}{81}z^{-4}$$

$$x(n) = \{\underset{\uparrow}{0}, \frac{1}{3}, \frac{4}{9}, \frac{13}{27}, \frac{52}{81}, \dots\}.$$

Module 5

Q19 A discrete system is given by the following difference equation: $y[n]-5\,y[n-1]=x[n]+4\,x[n-1]$ where, $x[n]$ is the input and $y[n]$ is the output. Determine the expression for its magnitude response and phase response.

Ans To determine the magnitude and phase response of the given discrete system, we can first take the z-transform of the difference equation:

Y(z) - 5z^-1 Y(z) = X(z) + 4z^-1 X(z)

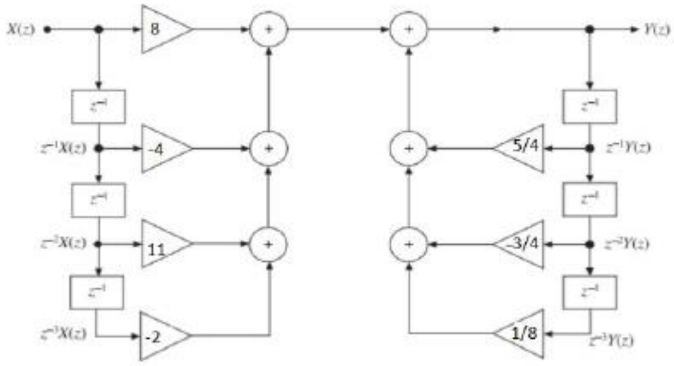
Simplifying this expression by factoring out Y(z) and X(z), we get:

Y(z) (1 - 5z^-1) = X(z) (1 + 4z^-1)

Dividing both sides by X(z), we get:

		Y(z)/X(z) = (1 + 4z^-1)/(1 - 5z^-1) The magnitude response of the system is given by the absolute value of the transfer function: |H(z)| = |Y(z)/X(z)| = |(1 + 4z^-1)/(1 - 5z^-1)| To find the phase response, we need to find the angle of the transfer function: ∠H(z) = ∠(Y(z)/X(z)) = ∠(1 + 4z^-1) - ∠(1 - 5z^-1) The angle of (1 + 4z^-1) is arctan(4/1) = 75.96 degrees, and the angle of (1 - 5z^-1) is arctan(-5/1) = -78.69 degrees. Therefore, the phase response is: ∠H(z) = 75.96 degrees - (-78.69 degrees) = 154.65 degrees So the expression for the magnitude response is |H(z)| = |(1 + 4z^-1)/(1 - 5z^-1)| and the expression for the phase response is ∠H(z) = 154.65 degrees.
	Q20	Obtain the direct form-I and direct form-II realizations of the discrete time system with system function as: $H(z) = \frac{8 - 4Z^{-1} + 11Z^{-2} - 2Z^{-3}}{1 - \frac{5}{4}Z^{-1} + \frac{3}{4}Z^{-2} - \frac{1}{8}Z^{-3}}$
	Ans	

Direct I form realization-4 marks



Direct II form realization-4 marks